

Figure 4: A triangle and square in Kturtle

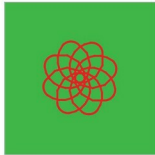


Figure 5: A flower with Kturtle

which has a triangle and a square with the green turtle placed near a corner of the square on the canvas. How will you send the turtle out of the canvas so that it does not obscure our view? Add the line of code `'go 1000, 1000'` at the end of the program to do this. This line of code sets the position of the turtle to location (1000, 1000) in the 2D plane of the canvas. The size of the canvas is only 500 x 500; so the turtle moves out of the visible area of the canvas.

Before understanding the program in detail, we need to understand the co-ordinate system of Kturtle. The co-ordinates of the top left corner are (0,0), the top right corner are (x,0), the bottom left corner are (0,y) and the bottom right corner are (x,y) where the values x and y represent the x-axis and the y-axis positions on the canvas. Now let us try to understand the program, line by line. The `'reset'` command resets everything done by the previous execution of a program and results in an empty canvas. We have already seen the purpose of the command `canvassize`. The command `'repeat'` is used to create a loop. So, the line of code `'repeat 3'` will execute all the lines of code inside the curly bracket three times. The line of code `'forward 100'` will draw a line of length 100 pixels. The width of the line drawn by default is 1 pixel. But the command `'penwidth'` can be used to modify the width of the line.

An important point to note is the direction in which the first line is drawn because we haven't specified the angle. An angle of zero degree means the line is drawn vertically. For a better understanding of this, type and execute the following two lines of code on the editor:

```
'revert'
```

...and:

```
'forward 100'.
```

You will see a horizontal line drawn on the canvas with the turtle resting at the top of the line. The line of code:

```
'turnleft 120'
```

...tells the turtle to turn an amount of 120 degrees to the left. Since the last two lines of code are repeated three times, three lines are drawn by the turtle to form a triangle. The `'go'` command tells the turtle to go to a certain place on the canvas. Since the line of code is `'go 400, 400'`, this place is 400 pixels from the left of the canvas and 400 pixels from the top of the canvas. After this, the following two lines of code:

```
'forward 100'
```

...and:

```
'turnleft 90'
```

...are repeated four times to draw a square.

Now let us go through another example, also provided by Kturtle, called 'flower'. A slightly modified version of the Kturtle program 'flower' is shown below.

```
reset
canvасcolor 0, 200, 0
pencolor 255, 0, 0
penwidth 3
repeat 8 {
  repeat 4 {
    forward 20
    turnright 90
  }
  repeat 7 {
    forward 10
    turnright 15
  }
  repeat 9 {
    forward 3
    turnright 10
  }
}
go 1000, 1000
```

There are only four lines of code that have been newly introduced in this program. The line of code `canvасcolor 0, 200, 0` is used to set the background colour of the canvas. You have to provide the red, green and blue components of the required colour as parameters to the command `'canvасcolor'`. Here, the background colour is green with an intensity 200. The line of code:

```
pencolor 255, 0, 0
```

...will fix the colour of the lines drawn by the turtle. Here again we use the RGB colour model. In this case, the colour of the lines drawn is red. The line of code:

```
penwidth 3
```